

```
> restart
```

Guessing examples with the package

DalgGuessing from

<https://github.com/T3gu1a/D-algebraic-Guessing>

```
>
>
> with(DalgGuessing)
                                [DalgFunGuess, modDalgFunGuess] (1)
> with(CodeTools, CPUTime) #CPUTime is often used to display the CPU time of computations
                                [CPUTime] (2)
>
```

▼ DalgFunGuess

```
>
```

▼ **Example 1:** $\zeta(2n+2), n=0, 1, 2 \dots$

```
>
> L := [seq(Zeta(2*j+2), j=0..12)] #13 terms
L := [  $\frac{\pi^2}{6}, \frac{\pi^4}{90}, \frac{\pi^6}{945}, \frac{\pi^8}{9450}, \frac{\pi^{10}}{93555}, \frac{691 \pi^{12}}{638512875}, \frac{2 \pi^{14}}{18243225}, \frac{3617 \pi^{16}}{325641566250},$  (1.1.1)
 $\frac{43867 \pi^{18}}{38979295480125}, \frac{174611 \pi^{20}}{1531329465290625}, \frac{155366 \pi^{22}}{13447856940643125},$ 
 $\frac{236364091 \pi^{24}}{201919571963756521875}, \frac{1315862 \pi^{26}}{11094481976030578125} ]$ 
```

```
> DalgFunGuess(L, degADE=2, degPoly=1, startfromord=2)

$$-C y(x)^2 + \frac{5_C \left( \frac{d}{dx} y(x) \right)}{2} - 2_C x \left( \frac{d}{dx} y(x) \right) y(x) + C x \left( \frac{d^2}{dx^2} y(x) \right) = 0 \quad (1.1.2)$$

```

```
> DalgFunGuess(L[1..9], degADE=2, degPoly=1, startfromord=2, allPolyDeg=true)
#9 terms

$$-C y(x)^2 + \frac{5_C \left( \frac{d}{dx} y(x) \right)}{2} - 2_C x \left( \frac{d}{dx} y(x) \right) y(x) + C x \left( \frac{d^2}{dx^2} y(x) \right) = 0 \quad (1.1.3)$$

```

```
> CPUTime(DalgFunGuess(L[1..7], degADE=2, degPoly=1, startfromord=2,
```

$allPolyDeg = true, sparsity = \frac{1}{2} \Big) \Big) \#7 \text{ terms}$

$$0.266, _C y(x)^2 + \frac{5_C \left(\frac{d}{dx} y(x) \right)}{2} - 2_C x \left(\frac{d}{dx} y(x) \right) y(x) + _C x \left(\frac{d^2}{dx^2} y(x) \right) = 0 \quad (1.1.4)$$

>
>

Example 2: Gregory coefficients $\frac{x}{\ln(1+x)} = 1 + \sum_{n=0}^{\infty} G_n x^n$

>
> $f := (x/\log(1+x) - 1)/x :$
> $L := PolynomialTools:-CoefficientList(convert(series(f, x, 21), polynom), x)$

$$L := \left[\frac{1}{2}, -\frac{1}{12}, \frac{1}{24}, -\frac{19}{720}, \frac{3}{160}, -\frac{863}{60480}, \frac{275}{24192}, -\frac{33953}{3628800}, \frac{8183}{1036800}, \right. \\ \left. -\frac{3250433}{479001600}, \frac{4671}{788480}, -\frac{13695779093}{2615348736000}, \frac{2224234463}{475517952000}, \right. \\ \left. -\frac{132282840127}{31384184832000}, \frac{2639651053}{689762304000}, -\frac{111956703448001}{32011868528640000}, \frac{50188465}{15613165568}, \right. \\ \left. -\frac{2334028946344463}{7860144949376000}, \frac{301124035185049}{109285437800448000} \right] \quad (1.2.1)$$

> $numelems(L)$

19 (1.2.2)

> $DalgFunGuess(L, degADE = 2, degPoly = 2, startfromord = 2) \#19 \text{ terms}$

$$_C y(x)^2 + 2_C x \left(\frac{d}{dx} y(x) \right) y(x) + (2_C x + 3_C) \left(\frac{d}{dx} y(x) \right) + (_C x^2 + _C x) \left(\frac{d^2}{dx^2} y(x) \right) = 0 \quad (1.2.3)$$

> $CPUTime(DalgFunGuess(L[1..11], degADE = 2, degPoly = 2, startfromord = 2, allPolyDeg = true)) \#11 \text{ terms}$

$$1.328, _C y(x)^2 + 2_C x \left(\frac{d}{dx} y(x) \right) y(x) + (2_C x + 3_C) \left(\frac{d}{dx} y(x) \right) + (_C x^2 + _C x) \left(\frac{d^2}{dx^2} y(x) \right) = 0 \quad (1.2.4)$$

> $CPUTime\left(DalgFunGuess\left(L[1..9], degADE = 2, degPoly = 2, startfromord = 2, allPolyDeg = true, sparsity = \frac{1}{2} \right) \right) \#9 \text{ terms and } \text{wrong guess}$

$$\begin{aligned}
28.656, & \frac{298462802951078}{6212024644107945} -C x^2 y(x)^2 + \left(\frac{75794151775174128}{414134976273863} -C x \right. \\
& + \left. \frac{347399047730767608}{2070674881369315} -C x^2 \right) \left(\frac{d}{dx} y(x) \right)^2 + \left(-\frac{91250717380829504}{2070674881369315} -C x \right. \\
& - \left. \frac{38269387337136158}{2070674881369315} -C x^2 \right) \left(\frac{d}{dx} y(x) \right) y(x) + \left(\frac{77205906414442392}{2070674881369315} -C x \right. \\
& + \left. \frac{33806828819378548}{6212024644107945} -C x^2 \right) \left(\frac{d}{dx} y(x) \right) + \left(-\frac{69769792316}{414134976273863} -C x \right. \\
& + \left. -C x^2 \right) \left(\frac{d^2}{dx^2} y(x) \right) = 0
\end{aligned} \tag{1.2.5}$$

> CPUTime(DalgFunGuess(L[1..10], degADE=2, degPoly=2, startfromord=2, allPolyDeg=true, sparsity=1/2)) #10 terms

$$\begin{aligned}
2.407, & -C y(x)^2 + 2 -C x \left(\frac{d}{dx} y(x) \right) y(x) + (2 -C x + 3 -C) \left(\frac{d}{dx} y(x) \right) + (-C x^2 \\
& + -C x) \left(\frac{d^2}{dx^2} y(x) \right) = 0
\end{aligned} \tag{1.2.6}$$

Example 3: Dividing D-algebraic functions:

An instance of finding an algebraic differential equation satisfied by

ALL $\frac{f}{g}$, where $p(f) = q(g) = 0$

```

1 #The differential polynomials
2 p:=diff(y(x), x) - x*y(x)^2:
3 q:=-diff(z(x), x) + z(x) + x + 1:
4 sys := {p=0,q=0}: #Note: it is better to treat both ADEs separately as they are independent
5
6 #symbolic initial conditions
7 InitialConds:= {y(0) = theta, z(0) = beta}:
8
9 #Order is a global variable for truncating series solutions
10 Order :=19:
11
12 #Computing truncations of f and g of order Order
13 sol := dsolve(sys union InitialConds {y(x), z(x)}, series):
14
15 # Extract the truncations
16 f_series := eval(y(x), sol):
17 g_series := eval(z(x), sol):
18
19 # Convert to polynomial
20 h_series := series(f_series/g_series, x, Order):
21 h_poly := convert(h_series, polynom):
22
23 # List of coefficients
24 L:=PolynomialTools-CoefficientList(h_poly,x):

```

> simplify(L[1..5]) #5 first terms

$$\left[\frac{\theta}{\beta}, -\frac{\theta(1+\beta)}{\beta^2}, \frac{(2+(\theta+1)\beta^2+2\beta)\theta}{2\beta^3}, \right. \tag{1.3.1}$$

$$- \frac{\left(2 + \left(\theta + \frac{1}{3}\right) \beta^3 + \left(\theta + \frac{2}{3}\right) \beta^2 + 2 \beta\right) \theta}{2 \beta^4},$$

$$\left[\frac{\left(4 + \left(\theta^2 + \theta + \frac{1}{6}\right) \beta^4 + \left(2 \theta - \frac{1}{3}\right) \beta^3 + \left(2 \theta + \frac{2}{3}\right) \beta^2 + 4 \beta\right) \theta}{4 \beta^5} \right]$$

> numelems(L)

19

(1.3.2)

> t, ADE := CPUTime(DalgFunGuess(L, degADE=2, degPoly=5, startfromord=1, inputConstants={beta, theta}))#19 terms

$$t, ADE := 1.672, \left(-\frac{2_C}{\theta^2} - \frac{2_C x}{\theta^2} + \frac{2_C x^2}{\theta} + \frac{2_C x^3}{\theta} - \frac{C x^4}{2} - \frac{C x^5}{2} \right) y(x)^2 + \left(-\frac{2_C}{\theta} + 2_C x + _C x^2 \right) y(x) + \left(-\frac{2_C}{\theta} + _C x^2 \right) \left(\frac{d}{dx} y(x) \right) = 0 \quad (1.3.3)$$

> CPUTime(DalgFunGuess(L[1..13], degADE=2, degPoly=5, startfromord=1, inputConstants={beta, theta}, allPolyDeg=true))#13 terms

$$6.250, \left(-\frac{2_C}{\theta^2} - \frac{2_C x}{\theta^2} + \frac{2_C x^2}{\theta} + \frac{2_C x^3}{\theta} - \frac{C x^4}{2} - \frac{C x^5}{2} \right) y(x)^2 + \left(-\frac{2_C}{\theta} + 2_C x + _C x^2 \right) y(x) + \left(-\frac{2_C}{\theta} + _C x^2 \right) \left(\frac{d}{dx} y(x) \right) = 0 \quad (1.3.4)$$

> CPUTime(DalgFunGuess(L[1..12], degADE=2, degPoly=5, startfromord=1, inputConstants={beta, theta}, allPolyDeg=true, sparsity=1/3, maxIteration=50))

#12 terms, would only succeed with luck

12.359, FAIL

(1.3.5)

Eliminating θ

> $\rho_1 := \text{subs}([_C = 2 \cdot \theta^2, y(x) = y, \text{diff}(y(x), x) = y1], \text{lhs}(ADE))$

$$\rho_1 := (-\theta^2 x^5 - \theta^2 x^4 + 4 \theta x^3 + 4 \theta x^2 - 4 x - 4) y^2 + (2 \theta^2 x^2 + 4 \theta^2 x - 4 \theta) y + (2 \theta^2 x^2 - 4 \theta) y1 \quad (1.3.6)$$

> $\rho_2 := \text{subs}([_C = 2 \cdot \theta^2, y(x) = y, \text{diff}(y(x), x) = y1, \text{diff}(y(x), x, x) = y2], \text{diff}(\text{lhs}(ADE), x))$

$$\rho_2 := (-5 \theta^2 x^4 - 4 \theta^2 x^3 + 12 \theta x^2 + 8 \theta x - 4) y^2 + 2 (-\theta^2 x^5 - \theta^2 x^4 + 4 \theta x^3 + 4 \theta x^2 - 4 x - 4) y y1 + (4 \theta^2 x + 4 \theta^2) y + (2 \theta^2 x^2 + 4 \theta^2 x - 4 \theta) y1 \quad (1.3.7)$$

$$+ 4 \theta^2 x y l + (2 \theta^2 x^2 - 4 \theta) y^2$$

> rho := resultant($\rho_1, \rho_2, \text{theta}$)

$$\begin{aligned} \rho := & -1024 x^6 y^7 - 3072 x^5 y^7 - 3072 x^4 y^7 - 256 x^4 y^6 - 1536 x^4 y^5 y l + 1024 x^4 y^5 y^2 \quad (1.3.8) \\ & - 2048 x^4 y^4 y l^2 - 1024 x^3 y^7 - 1536 x^3 y^6 - 4096 x^3 y^5 y l + 2048 x^3 y^5 y^2 \\ & - 4096 x^3 y^4 y l^2 - 1280 x^2 y^6 - 3072 x^2 y^5 y l + 1024 x^2 y^5 y^2 - 2048 x^2 y^4 y l^2 \\ & - 256 x^2 y^4 y l + 256 x^2 y^4 y^2 - 1024 x^2 y^3 y l^2 + 768 x^2 y^3 y l y^2 - 256 x^2 y^3 y^2^2 \\ & - 1536 x^2 y^2 y l^3 + 1024 x^2 y^2 y l^2 y^2 - 1024 x^2 y y l^4 - 512 x y^5 y l - 256 x y^5 \\ & - 1280 x y^4 y l + 768 x y^4 y^2 - 2816 x y^3 y l^2 + 1280 x y^3 y l y^2 - 256 x y^3 y^2^2 \\ & - 2560 x y^2 y l^3 + 1024 x y^2 y l^2 y^2 - 1024 x y y l^4 + 256 y^6 - 256 y^5 - 768 y^4 y l \\ & + 256 y^4 y^2 - 1024 y^3 y l^2 + 256 y^3 y l y^2 - 512 y^2 y l^3 \end{aligned}$$

> factor(rho)

$$\begin{aligned} & -256 y (4 x^6 y^6 + 12 x^5 y^6 + 12 x^4 y^6 + x^4 y^5 + 6 x^4 y^4 y l - 4 x^4 y^4 y^2 + 8 x^4 y^3 y l^2 \quad (1.3.9) \\ & + 4 x^3 y^6 + 6 x^3 y^5 + 16 x^3 y^4 y l - 8 x^3 y^4 y^2 + 16 x^3 y^3 y l^2 + 5 x^2 y^5 + 12 x^2 y^4 y l \\ & - 4 x^2 y^4 y^2 + 8 x^2 y^3 y l^2 + x^2 y^3 y l - x^2 y^3 y^2 + 4 x^2 y^2 y l^2 - 3 x^2 y^2 y l y^2 \\ & + x^2 y^2 y^2^2 + 6 x^2 y y l^3 - 4 x^2 y y l^2 y^2 + 4 x^2 y l^4 + 2 x y^4 y l + y^4 x + 5 x y^3 y l \\ & - 3 x y^3 y^2 + 11 x y^2 y l^2 - 5 x y^2 y l y^2 + x y^2 y^2^2 + 10 x y y l^3 - 4 x y y l^2 y^2 \\ & + 4 x y l^4 - y^5 + y^4 + 3 y^3 y l - y^3 y^2 + 4 y^2 y l^2 - y^2 y l y^2 + 2 y y l^3) \end{aligned}$$

> rho := (collect(op(3, (1.3.9)), [y, y l, y^2], distributed))

$$\begin{aligned} \rho := & (4 x^6 + 12 x^5 + 12 x^4 + 4 x^3) y^6 + (x^4 + 6 x^3 + 5 x^2 - 1) y^5 + (6 x^4 + 16 x^3 \quad (1.3.10) \\ & + 12 x^2 + 2 x) y^4 y l + (-4 x^4 - 8 x^3 - 4 x^2) y^4 y^2 + (1 + x) y^4 + (8 x^4 + 16 x^3 \\ & + 8 x^2) y^3 y l^2 + (x^2 + 5 x + 3) y^3 y l + (-x^2 - 3 x - 1) y^3 y^2 + (4 x^2 + 11 x \\ & + 4) y^2 y l^2 + (-3 x^2 - 5 x - 1) y^2 y l y^2 + (x^2 + x) y^2 y^2^2 + (6 x^2 + 10 x \\ & + 2) y y l^3 + (-4 x^2 - 4 x) y y l^2 y^2 + (4 x^2 + 4 x) y l^4 \end{aligned}$$

> ADE := subs([y=y(x), y l=diff(y(x), x), y^2=diff(y(x), x, x)], rho)

$$\begin{aligned} ADE := & (4 x^6 + 12 x^5 + 12 x^4 + 4 x^3) y(x)^6 + (x^4 + 6 x^3 + 5 x^2 - 1) y(x)^5 + (6 x^4 \quad (1.3.11) \\ & + 16 x^3 + 12 x^2 + 2 x) y(x)^4 \left(\frac{d}{dx} y(x) \right) + (-4 x^4 - 8 x^3 - 4 x^2) y(x)^4 \left(\frac{d^2}{dx^2} \right. \\ & \left. y(x) \right) + (1 + x) y(x)^4 + (8 x^4 + 16 x^3 + 8 x^2) y(x)^3 \left(\frac{d}{dx} y(x) \right)^2 + (x^2 + 5 x \\ & + 3) y(x)^3 \left(\frac{d}{dx} y(x) \right) + (-x^2 - 3 x - 1) y(x)^3 \left(\frac{d^2}{dx^2} y(x) \right) + (4 x^2 + 11 x \\ & + 4) y(x)^2 \left(\frac{d}{dx} y(x) \right)^2 + (-3 x^2 - 5 x - 1) y(x)^2 \left(\frac{d}{dx} y(x) \right) \left(\frac{d^2}{dx^2} y(x) \right) \\ & + (x^2 + x) y(x)^2 \left(\frac{d^2}{dx^2} y(x) \right)^2 + (6 x^2 + 10 x + 2) y(x) \left(\frac{d}{dx} y(x) \right)^3 + (\end{aligned}$$

$$-4x^2 - 4x) y(x) \left(\frac{d}{dx} y(x) \right)^2 \left(\frac{d^2}{dx^2} y(x) \right) + (4x^2 + 4x) \left(\frac{d}{dx} y(x) \right)^4$$

NLDE finds the same ADE from p and q.

$$\begin{aligned} & \text{CPUTime} \left(\text{NLDE:-arithmeticDalg} \left([p=0, q=0], [y(x), z(x)], w = \frac{y}{z} \right) \right) \\ 2.266, & (4x^6 + 12x^5 + 12x^4 + 4x^3) w(x)^6 + (x^4 + 6x^3 + 5x^2 - 1) w(x)^5 + (6x^4 \\ & + 16x^3 + 12x^2 + 2x) w(x)^4 \left(\frac{d}{dx} w(x) \right) + (-4x^4 - 8x^3 - 4x^2) \left(\frac{d^2}{dx^2} \right. \\ & w(x) \left. \right) w(x)^4 + (x + 1) w(x)^4 + (8x^4 + 16x^3 + 8x^2) w(x)^3 \left(\frac{d}{dx} w(x) \right)^2 \\ & + (x^2 + 5x + 3) w(x)^3 \left(\frac{d}{dx} w(x) \right) + (-x^2 - 3x - 1) w(x)^3 \left(\frac{d^2}{dx^2} w(x) \right) \\ & + (4x^2 + 11x + 4) w(x)^2 \left(\frac{d}{dx} w(x) \right)^2 + (-3x^2 - 5x - 1) w(x)^2 \left(\frac{d}{dx} \right. \\ & w(x) \left. \right) \left(\frac{d^2}{dx^2} w(x) \right) + (x^2 + x) w(x)^2 \left(\frac{d^2}{dx^2} w(x) \right)^2 + (6x^2 + 10x \\ & + 2) w(x) \left(\frac{d}{dx} w(x) \right)^3 + (-4x^2 - 4x) w(x) \left(\frac{d}{dx} w(x) \right)^2 \left(\frac{d^2}{dx^2} w(x) \right) \\ & + (4x^2 + 4x) \left(\frac{d}{dx} w(x) \right)^4 = 0 \end{aligned} \quad (1.3.12)$$

Example 4: ADE for $e^{\exp(x) - e}$

$$\begin{aligned} & f := \exp(\exp(\exp(x)) - \exp(1)) : \\ & L := \text{PolynomialTools:-CoefficientList}(\text{convert}(\text{series}(f, x, 50), \text{polynom}), x) : \\ & \text{numelems}(L) \\ & \qquad \qquad \qquad 50 \end{aligned} \quad (1.4.1)$$

$$\begin{aligned} & L[1..5] \\ & \left[1, e, e + \frac{(e)^2}{2}, \frac{5e}{6} + (e)^2 + \frac{(e)^3}{6}, \frac{5e}{8} + \frac{4(e)^2}{3} + \frac{(e)^3}{2} + \frac{(e)^4}{24} \right] \end{aligned} \quad (1.4.2)$$

$$\text{CPUTime}(\text{DalgFunGuess}(L, \text{degADE}=4, \text{degPoly}=0, \text{startfromord}=3, \text{linsolver} = \text{HardSystem}))$$

$$\text{CPUTime} \left(\text{DalgFunGuess} \left(L[1..20], \text{degADE}=4, \text{degPoly}=0, \text{startfromord}=4, \right. \right.$$

```
allPolyDeg = true, sparsity =  $\frac{1}{2}$  ) )
```

modDalgFunGuess

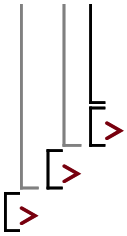
Example 5: OEIS [A189281](#)

```
> T := readdata("b189281.txt", integer, 2) :
> L := map(t -> t[2], T) :
> numelems(L)
301 (2.1.1)
```

```
> L[-1] #the last term in the list
11372069544670615182450141371651349899813311450244396383365131233787919\ (2.1.2)
03963787297097950914614622744980916245525095837059122625005087256009\
61267583819951354960326626583396441011329385038320608019808450146379\
81363927893371454163853400798548859033415579591422667854989517752461\
11392830369578261524732922126589677966923150803305798073113073143330\
01266371294938427626025774479608026492895343733562842729487316487770\
77733286230349235354762502184041098538039037527828777121056595891273\
03659194955994905489080540135700897583959649922532184971527954332600\
81844544920264233182418342148466552185999146775891138264272876438194
```

```
> CPUtime(modDalgFunGuess(L, degADE = 1, degPoly = 11, startfromord = 19, modulus
= 569))
4.890, (132 _C x^2 + 43 _C x + 266 _C)  $\left( \frac{d^2}{dx^2} y(x) \right) + (267 _C x^3 + 62 _C x^2$  (2.1.3)
+ 309 _C x + 50 _C)  $\left( \frac{d^3}{dx^3} y(x) \right) + (138 _C x^4 + 203 _C x^3 + 544 _C x^2$ 
+ 329 _C x + 436 _C)  $\left( \frac{d^4}{dx^4} y(x) \right) + (118 _C x^5 + 181 _C x^4 + 63 _C x^3$ 
+ 288 _C x^2 + 381 _C x + 444 _C)  $\left( \frac{d^5}{dx^5} y(x) \right) + (555 _C x^6 + 334 _C x^5$ 
+ 475 _C x^4 + 92 _C x^3 + 15 _C x^2 + 37 _C x + 211 _C)  $\left( \frac{d^6}{dx^6} y(x) \right)$ 
+ (195 _C x^7 + 524 _C x^6 + 294 _C x^5 + 69 _C x^4 + 459 _C x^3 + 179 _C x^2
```

$$\begin{aligned}
& + 56_C x + 195_C) \left(\frac{d^7}{dx^7} y(x) \right) + (337_C x^{11} + 165_C x^{10} + 256_C x^9 \\
& + 324_C x^8 + 473_C x^7 + 441_C x^6 + 54_C x^5 + 189_C x^4 + 357_C x^3 \\
& + 395_C x^2 + 381_C x + 90_C) \left(\frac{d^{13}}{dx^{13}} y(x) \right) + (513_C x^{11} + 354_C x^{10} \\
& + 379_C x^9 + 369_C x^8 + 306_C x^7 + 36_C x^6 + 242_C x^5 + 307_C x^4 \\
& + 522_C x^3 + 222_C x^2 + 108_C x + 68_C) \left(\frac{d^{14}}{dx^{14}} y(x) \right) + (518_C x^{11} \\
& + 410_C x^{10} + 35_C x^9 + 209_C x^8 + 23_C x^7 + 549_C x^6 + 333_C x^5 \\
& + 226_C x^4 + 210_C x^3 + 509_C x^2 + 186_C x + 188_C) \left(\frac{d^{15}}{dx^{15}} y(x) \right) \\
& + (553_C x^{11} + 544_C x^{10} + 557_C x^9 + 229_C x^8 + 315_C x^7 + 331_C x^6 \\
& + 376_C x^5 + 369_C x^4 + 191_C x^3 + 6_C x^2 + 441_C x + 4_C) \left(\frac{d^{16}}{dx^{16}} \right. \\
& \left. y(x) \right) + (412_C x^8 + 124_C x^7 + 340_C x^6 + 464_C x^5 + 237_C x^4 \\
& + 531_C x^3 + 351_C x^2 + 283_C x + 514_C) \left(\frac{d^8}{dx^8} y(x) \right) + (262_C x^9 \\
& + 535_C x^8 + 80_C x^7 + 233_C x^6 + 248_C x^5 + 565_C x^4 + 353_C x^3 \\
& + 380_C x^2 + 58_C x + 426_C) \left(\frac{d^9}{dx^9} y(x) \right) + (522_C x^{10} + 85_C x^9 \\
& + 21_C x^8 + 136_C x^7 + 58_C x^6 + 522_C x^5 + 263_C x^4 + 373_C x^3 \\
& + 565_C x + 233_C) \left(\frac{d^{10}}{dx^{10}} y(x) \right) + (447_C x^{11} + 458_C x^{10} + 57_C x^9 \\
& + 459_C x^8 + 320_C x^7 + 363_C x^6 + 479_C x^5 + 451_C x^4 + 445_C x^3 \\
& + 362_C x^2 + 61_C x + 545_C) \left(\frac{d^{11}}{dx^{11}} y(x) \right) + (417_C x^{11} + 375_C x^{10} \\
& + 63_C x^9 + 394_C x^8 + 35_C x^7 + 196_C x^6 + 195_C x^5 + 426_C x^4 \\
& + 39_C x^3 + 230_C x^2 + 154_C x + 321_C) \left(\frac{d^{12}}{dx^{12}} y(x) \right) + (523_C x^{11} \\
& + 375_C x^{10} + 47_C x^9 + 280_C x^8 + 338_C x^7 + 507_C x^6 + 278_C x^5 \\
& + 233_C x^4 + 565_C x^2) \left(\frac{d^{17}}{dx^{17}} y(x) \right) + (282_C x^{11} + 467_C x^{10} + 458_C x^9 \\
& + 135_C x^8 + 432_C x^7 + 566_C x^6 + 565_C x^5) \left(\frac{d^{18}}{dx^{18}} y(x) \right) + (_C x^{11}
\end{aligned}$$



$$+_C x^{10} + 568 _C x^9 + 568 _C x^8) \left(\frac{d^{19}}{dx^{19}} y(x) \right) = 0$$